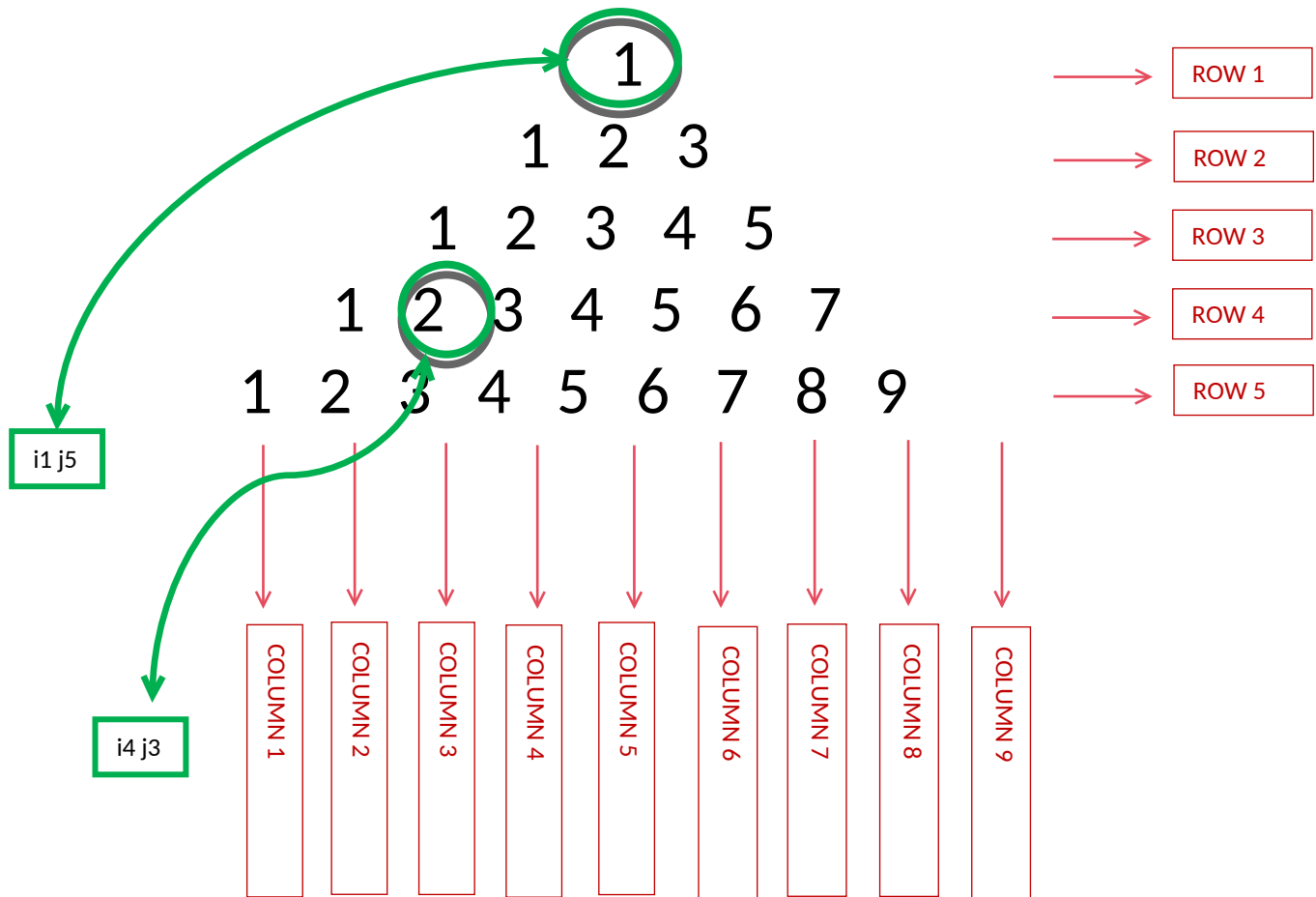


WEEK 01 MODULE
Programming Practice

TOPIC : 2 'Number Pattern'

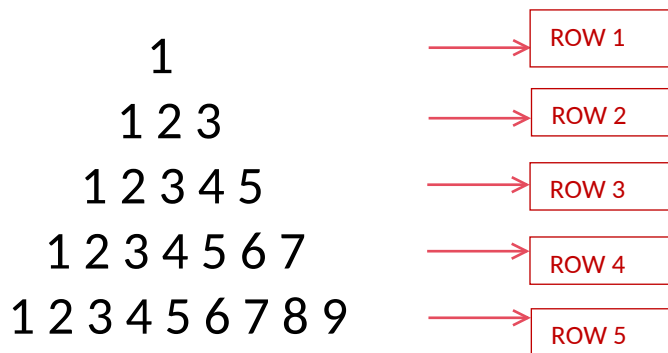
9. NUMBER PYRAMID



1. Problem Analysis :-

i. Pattern Structure:

- The output forms a centered number pyramid.
- Unlike the Triangle and Inverted Triangle patterns, the number are not aligned to the left side this is aligned around a central vertical axis.
- spaces are printed before the number so that the pattern appears centered.
- For $n=5$



ii. Execution Flow:

Step 1: Observe the Grid Concept:

- For $n = 5$

1
1 2 3
1 2 3 4 5
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8 9

The pattern grows symmetrically from the center.

As we move downward:

- The number of spaces decreases.
- The number increases.

Both of these changes occur simultaneously.

Therefore, this pattern requires two different calculations in every row.

- Each cell in the grid represents a unique coordinate (i, j) , where i denotes the row index and j denotes the column index.

-The pattern can be viewed as two regions:

✦ Region 1: Leading Spaces

These spaces shift the stars toward the center.

✦ Region 2: Stars

These form the visible pyramid.

For $n = 5$

Row 1 : [][1]

Row 2 : [][1 2 3]

Row 3 : [][1 2 3 4 5]

Row 4 : [][1 2 3 4 5 6 7]

Row 5 : [][1 2 3 4 5 6 7 8 9]

As the row index increases:

- Spaces decrease by 1.
- number increase by 2.

-> Row Analysis :-

Row Index (I)	1	2	3	4	5
Spaces	4	3	2	1	0
number	1	3	5	7	9
Printed value	1	123	12345	1234567	123456789

The pyramid is controlled by two formulas:

Spaces = $n - i - 1$

Number = $2 \times i + 1$

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Step 3: For every row, calculate the number of leading spaces required.

- The number of spaces depends on the current row index.
- The relationship is:

- Spaces = $n - i - 1$

Step 4: Traverse the required number of positions and print number.

- This shifts the number toward the center.
- The number of spaces decreases as the row index increases.

Step 5: After printing spaces, calculate the number required.

- The relationship is:

- number = $2 \times i + 1$

Step 6: Traverse the required number of positions and print numbers.

- The number increases by 2 in every row.

Step 7: After all number of the current row have been printed, move the cursor to the next line using `System.out.println();`

- This ensures that the next row starts on a new line.

Step 8: Repeat the process for all rows until the outer loop terminates.

2. Condition Analysis

No conditional statements are required.

There is no need for `if(...)`

- Because every position visited by the space loop prints a space.
- Every position visited by the number loop prints a number.
- The pattern is controlled entirely through loop limits.

3. Core Logic Transformation

Pyramid Pattern:-

```
for(int j = 0; j < 2 * i + 1; j++)
```

- Meaning:

numbers increase by 2 in every row.

Additionally :-

```
for(int j = 0; j < n - i - 1; j++)
```

- controls the alignment of the pattern.

4. Program :-

```
package week1_module;
import java.util.Scanner;
public class NumberPyramid {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 0; i < n; i++) {

            // Print spaces
            for(int j = 0; j < n - i - 1; j++) {
                System.out.print(" ");
            }

            // Print number
            for(int j = 1; j <= 2 * i + 1; j++) {
                System.out.print(j);
            }

            System.out.println();
        }

        sc.close();
    }
}
```

For pyramids, using **0-based indexing** ($i = 0$) ie formulas become:

Spaces = $n - i - 1$
Numbers = $2 * i + 1$

If you force 1-based indexing, the formulas become :

Spaces = $n - i$
Numbers = $2 * i - 1$

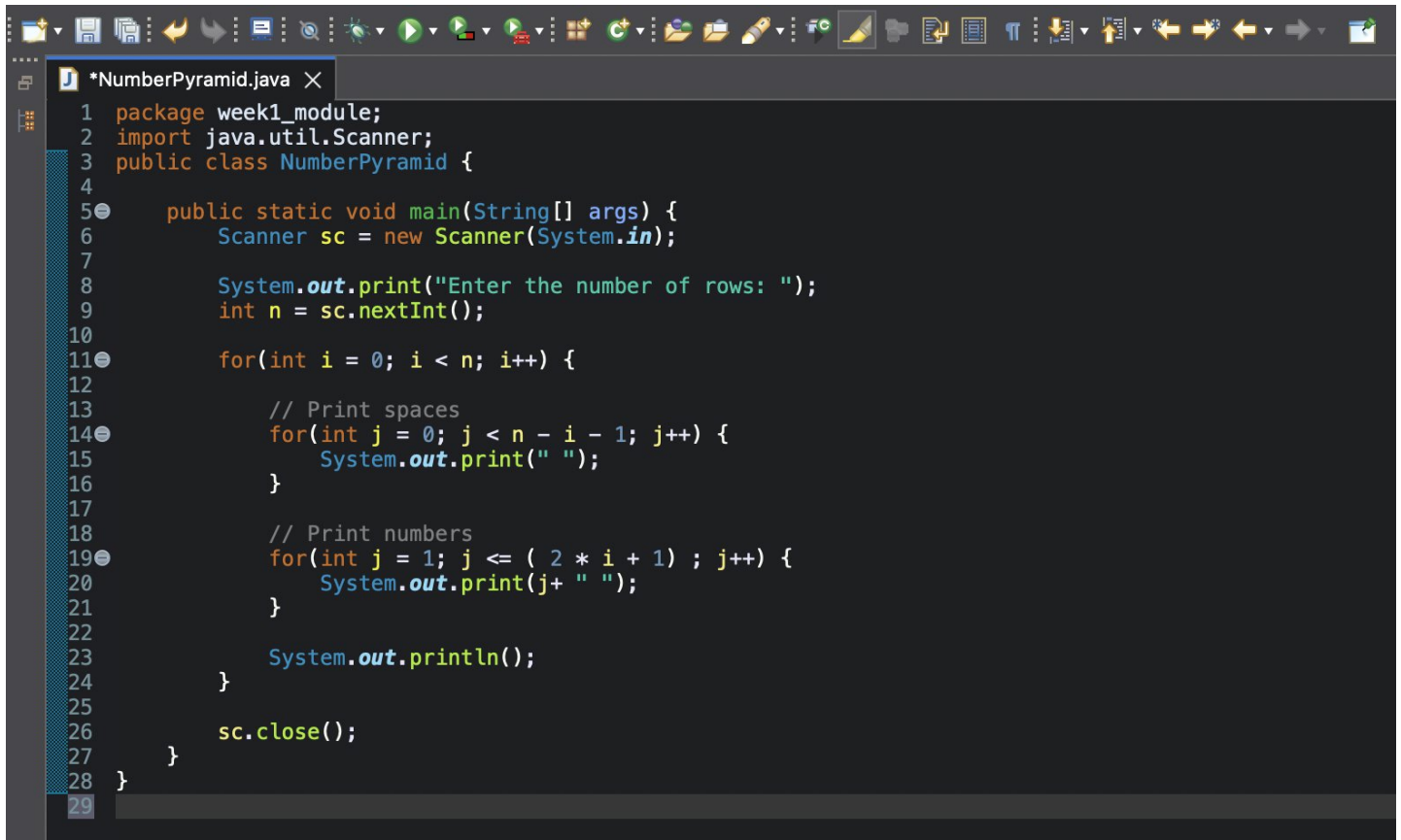
5. Program Output :-

Enter the number of rows: 5

```
      1
    1 2 3
  1 2 3 4 5
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8 9
```

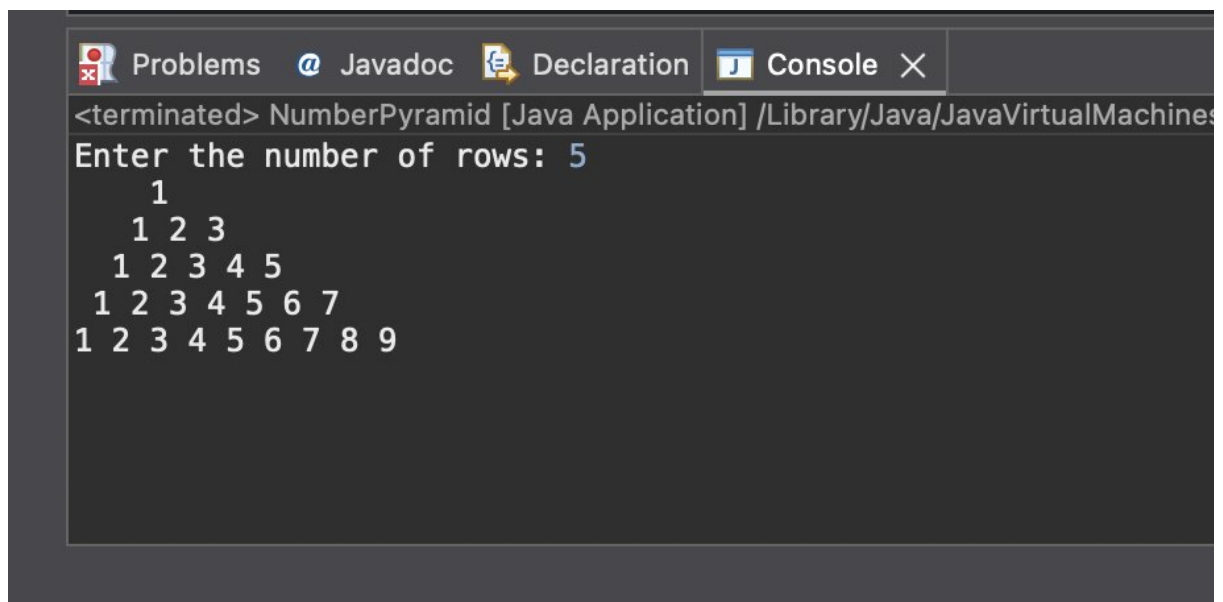
6. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class NumberPyramid {
4
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the number of rows: ");
9         int n = sc.nextInt();
10
11         for(int i = 0; i < n; i++) {
12             // Print spaces
13             for(int j = 0; j < n - i - 1; j++) {
14                 System.out.print(" ");
15             }
16
17             // Print numbers
18             for(int j = 1; j <= ( 2 * i + 1 ); j++) {
19                 System.out.print(j+ " ");
20             }
21
22             System.out.println();
23         }
24
25         sc.close();
26     }
27 }
28
29
```

ii. Program's Output :-



```
<terminated> NumberPyramid [Java Application] /Library/Java/JavaVirtualMachines
Enter the number of rows: 5
1
1 2 3
1 2 3 4 5
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8 9
```